

Physical Chemistry of Materials

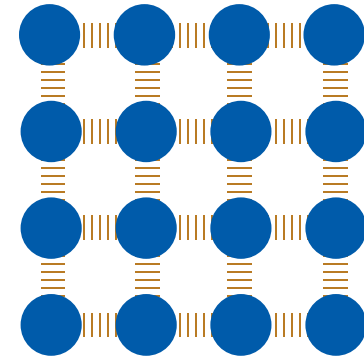
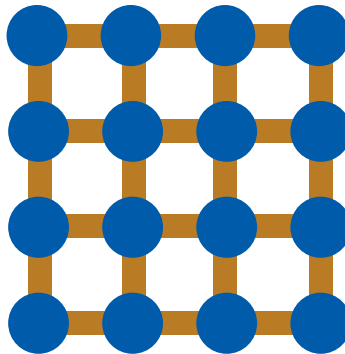
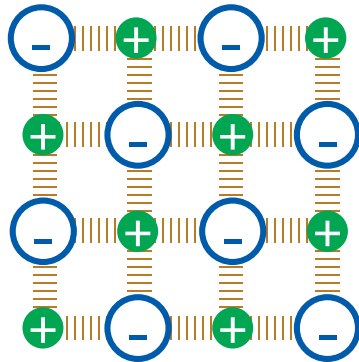
Lecture 11

Crystal Vibrations: Classical Picture



Bonding in Solids

- Ionic solids: ionic bonds
- Covalent solids: covalent bonds
- Molecular solids: intermolecular interactions



Metals need more discussion

Lattice Energy of Ionic Solids

- Lattice energy:

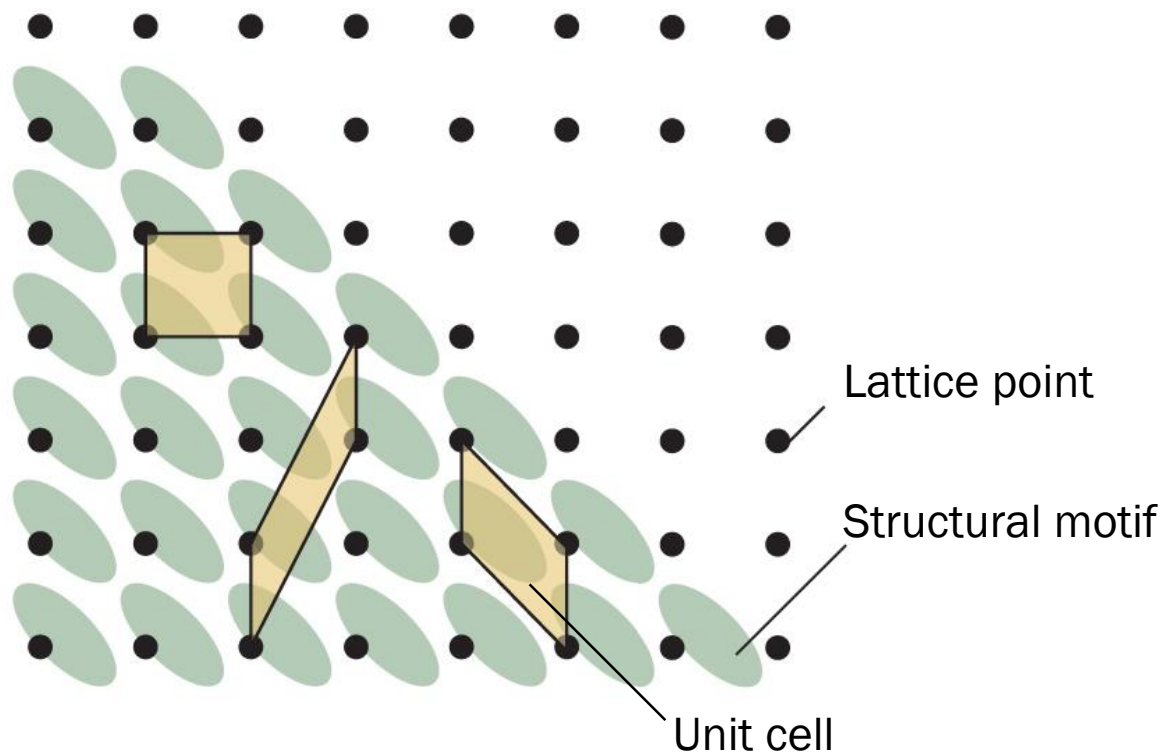


- Two main contributions
 - Electrostatic interactions (attraction + repulsion)
 - Pauli repulsion
- Born-Meyer equation

$$E_{\text{p,tot}}(d_{\text{eq}}) = -A \frac{|z_A z_B| e^2}{4\pi \epsilon_0 d_{\text{eq}}} \left(1 - \frac{d^*}{d_{\text{eq}}} \right)$$

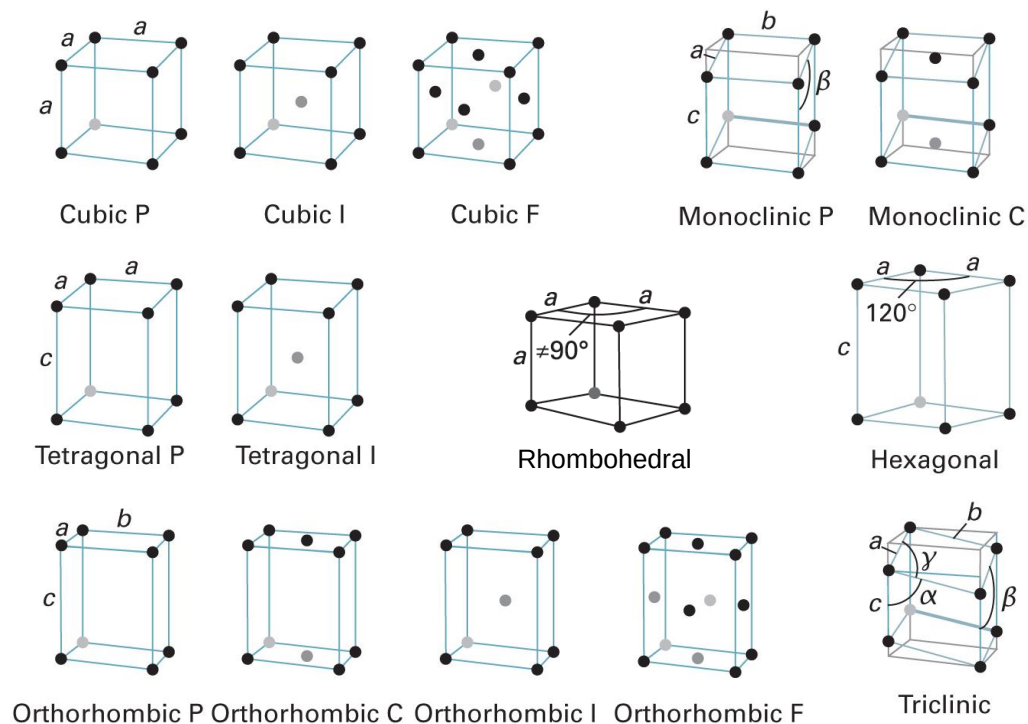
Unit Cells

whose purely translational displacements can span the entire space

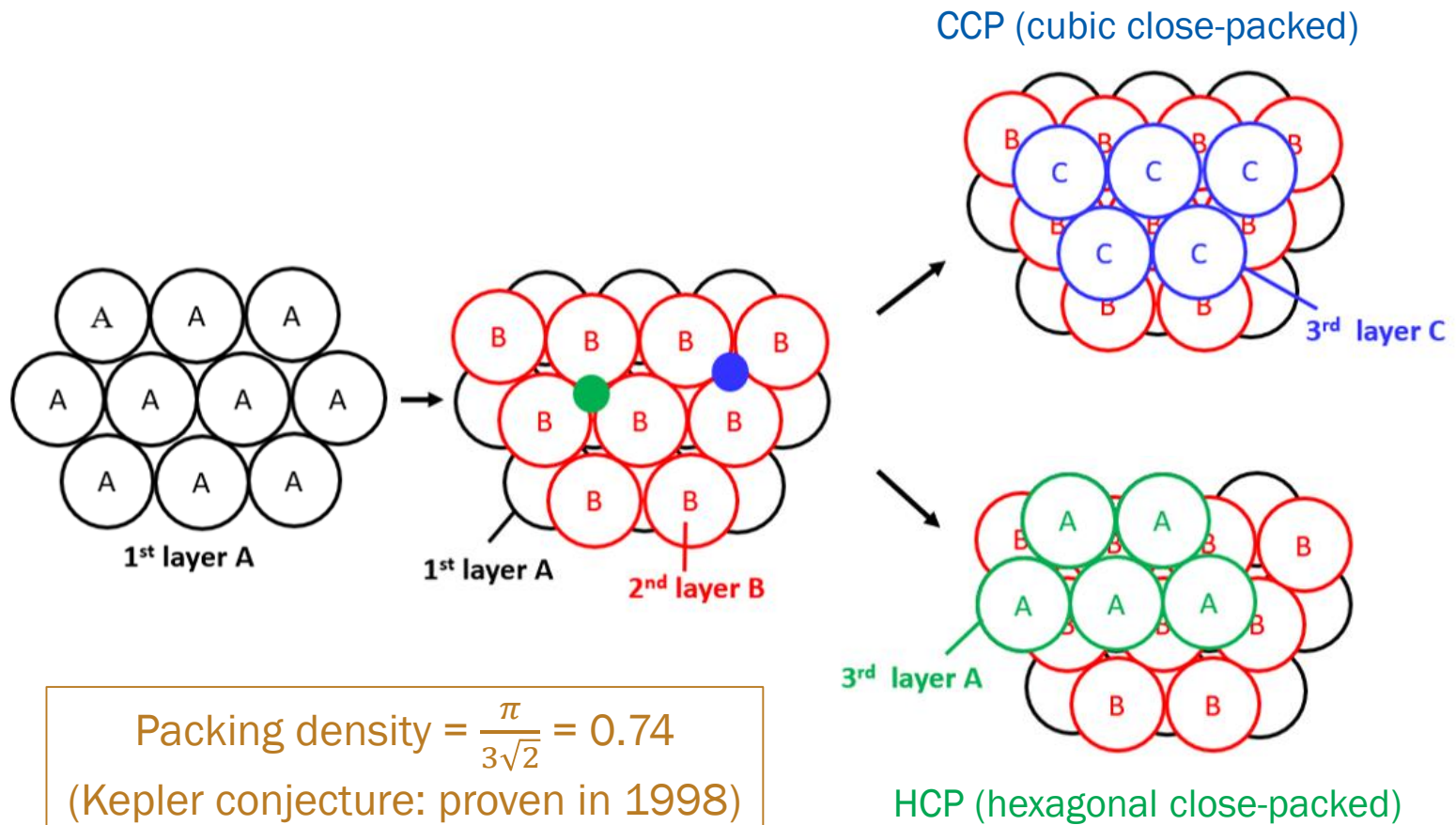


(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 15A.1 and 15A.2)

14 Bravais Lattices



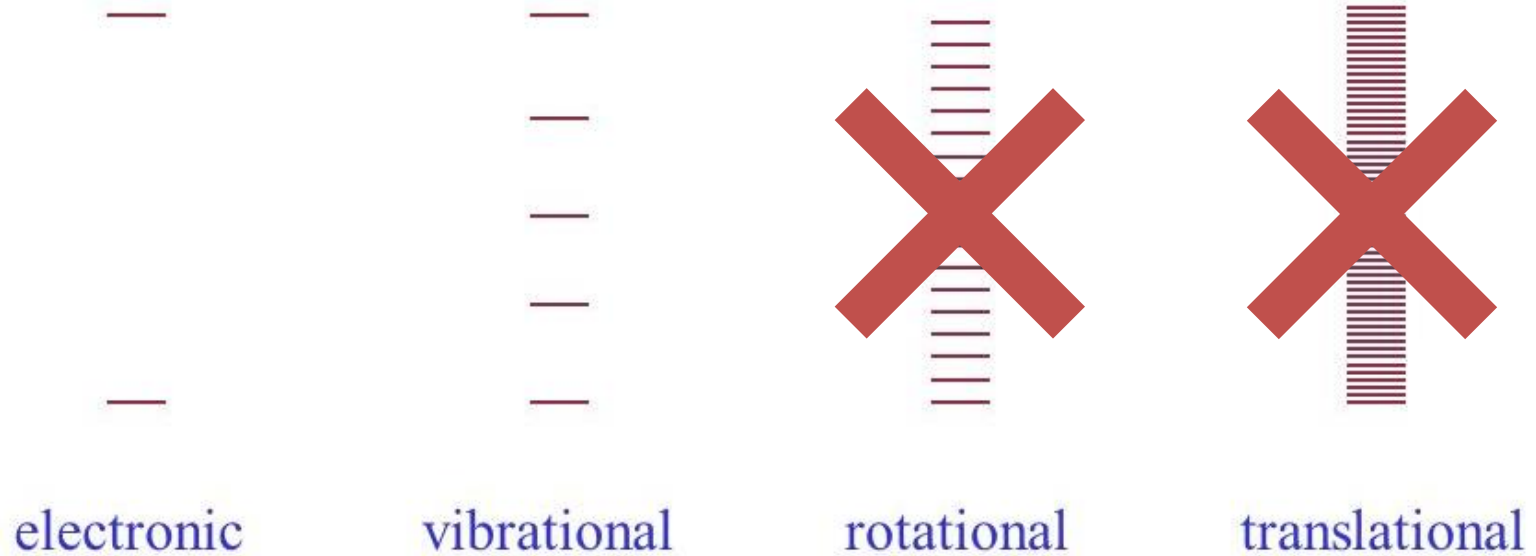
Close Packing



Last Class

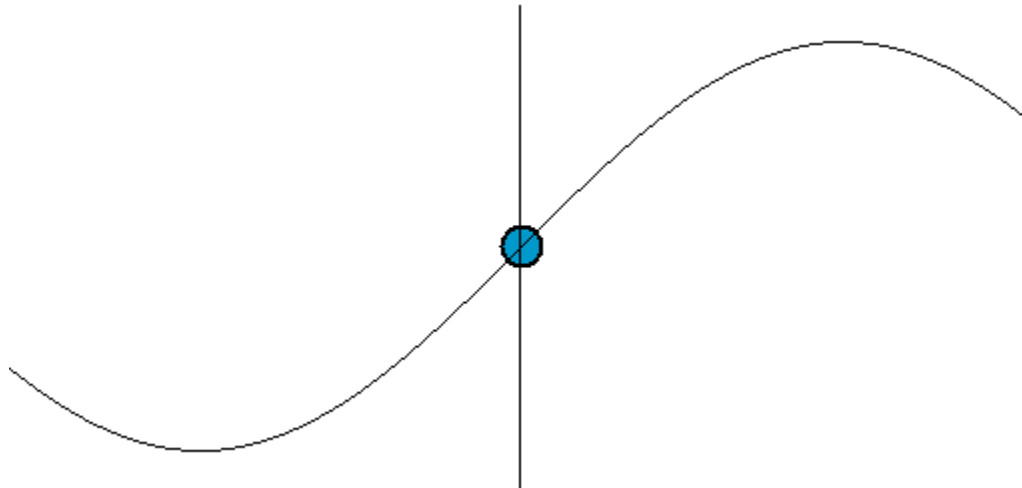
Crystal Energies

Molecular case:



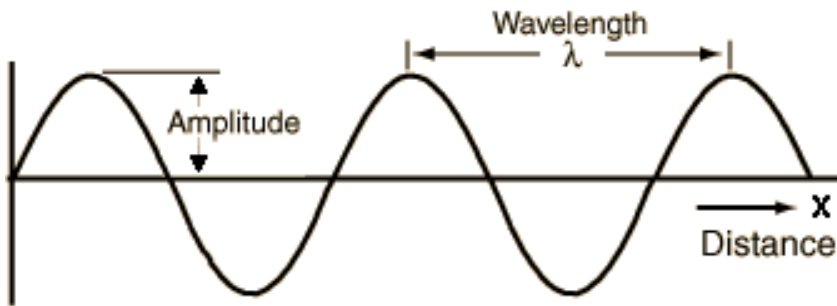
Waves

Temporal change + Spatial change

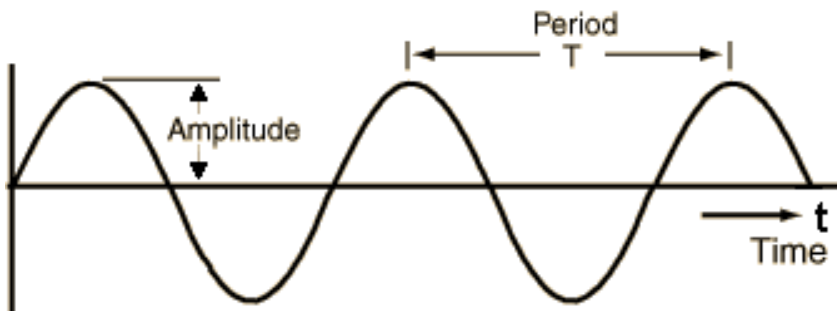


(Image: https://commons.wikimedia.org/wiki/File:Simple_harmonic_motion_animation.gif)

Basic Terminology



Wavenumber $\tilde{\nu} = 1/\lambda$

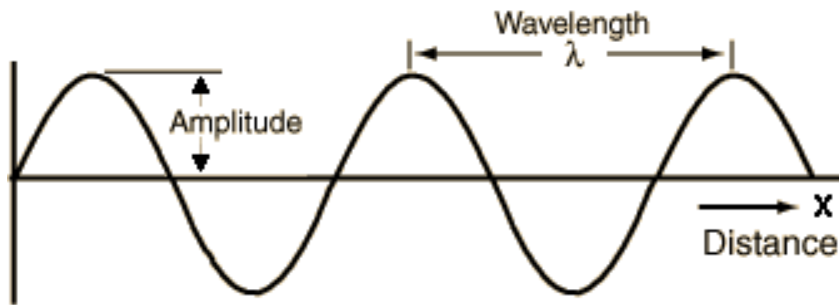


Frequency $\nu = 1/T$

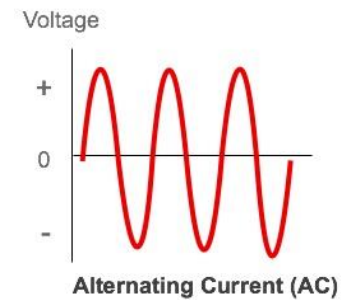
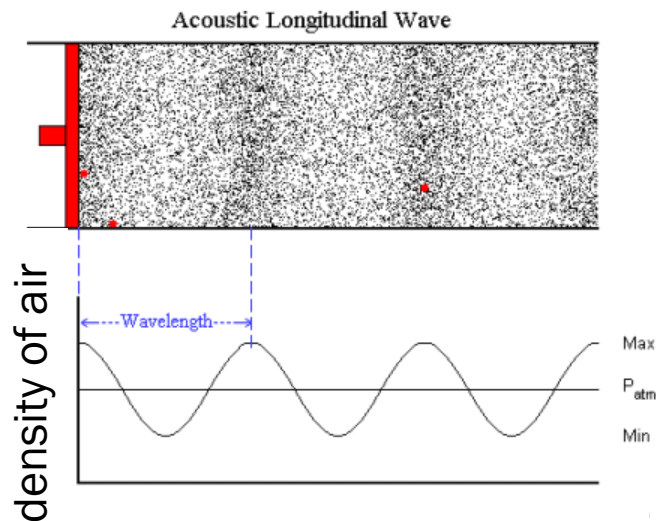
Speed $u = \lambda/T = \lambda\nu$

“Amplitude”

Some value assigned for each point of the space

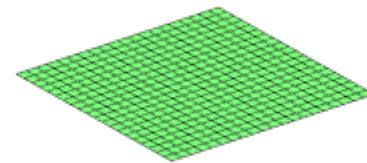
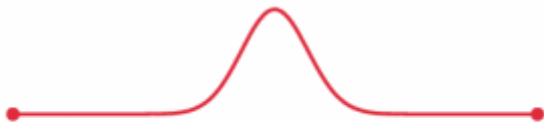


0 +1 +2 +1 0 -1 -2 -1 0 +1 +2 +1



Wave Equation

$$\frac{\partial^2 A}{\partial t^2} = u^2 \nabla^2 A$$



1D Traveling Wave



$$A(x, t) = A_0 \exp(ikx - i\omega t)$$

Wave equation: $\frac{\partial^2 A}{\partial t^2} = u^2 \frac{\partial^2 A}{\partial x^2}$

Substitution leads to

$$-A_0 \omega^2 \exp(ikx - i\omega t) = -A_0 u^2 k^2 \exp(ikx - i\omega t)$$

Hence, the wave equation is **satisfied** with $u = \omega/k$



1D Traveling Wave



$$A(x, t) = A_0 \exp(ikx - i\omega t)$$

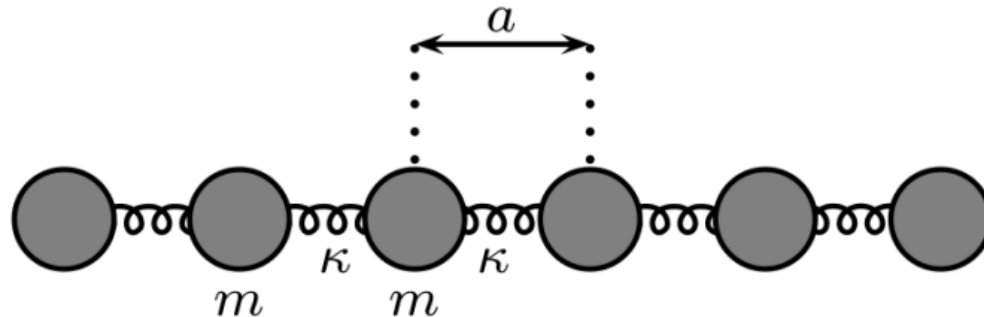
- Period: $\omega t = 2\pi$

$$T = 2\pi / \omega$$

- Wavelength: $kx = 2\pi$

$$\lambda = 2\pi / k$$

Normal Modes of 1D Crystal



- Deviation from equilibrium position $q_n = x_n - x_{n0}$
- Total potential energy

$$V_{\text{tot}} = \sum_{i=1}^N \frac{\kappa}{2} (x_{i+1} - x_i - a)^2 = \sum_{i=1}^N \frac{\kappa}{2} (q_{i+1} - q_i)^2$$

Newton's Law

$$V_{\text{tot}} = \sum_{i=1}^N \frac{\kappa}{2} (x_{i+1} - x_i - a)^2 = \sum_{i=1}^N \frac{\kappa}{2} (q_{i+1} - q_i)^2$$

- Force on particle n

$$F_n = -\frac{\partial V_{\text{tot}}}{\partial x_n} = \kappa(q_{n+1} - q_n) + \kappa(q_{n-1} - q_n)$$

$$F_n = ma_n = m \frac{\partial^2 q_n}{\partial t^2} = \kappa(q_{n+1} + q_{n-1} - 2q_n)$$

Solving the Differential Equation

$$m \frac{\partial^2 q_n}{\partial t^2} = \kappa (q_{n+1} + q_{n-1} - 2q_n)$$

- Ansatz (“educated guess”):

$$q_n = A \exp(i\omega t - ikx_{n0}) = A \exp(i\omega t - ikna)$$

Substitution leads to

$$-m\omega^2 A e^{i\omega t - ikna} = \kappa A (e^{i\omega t - ik(n+1)a} + e^{i\omega t - ik(n-1)a} - 2e^{i\omega t - ikna})$$

$$-m\omega^2 = \kappa (e^{-ika} + e^{ika} - 2)$$

$$m\omega^2 = 4\kappa \sin^2(ka/2)$$

Normal Mode Frequency

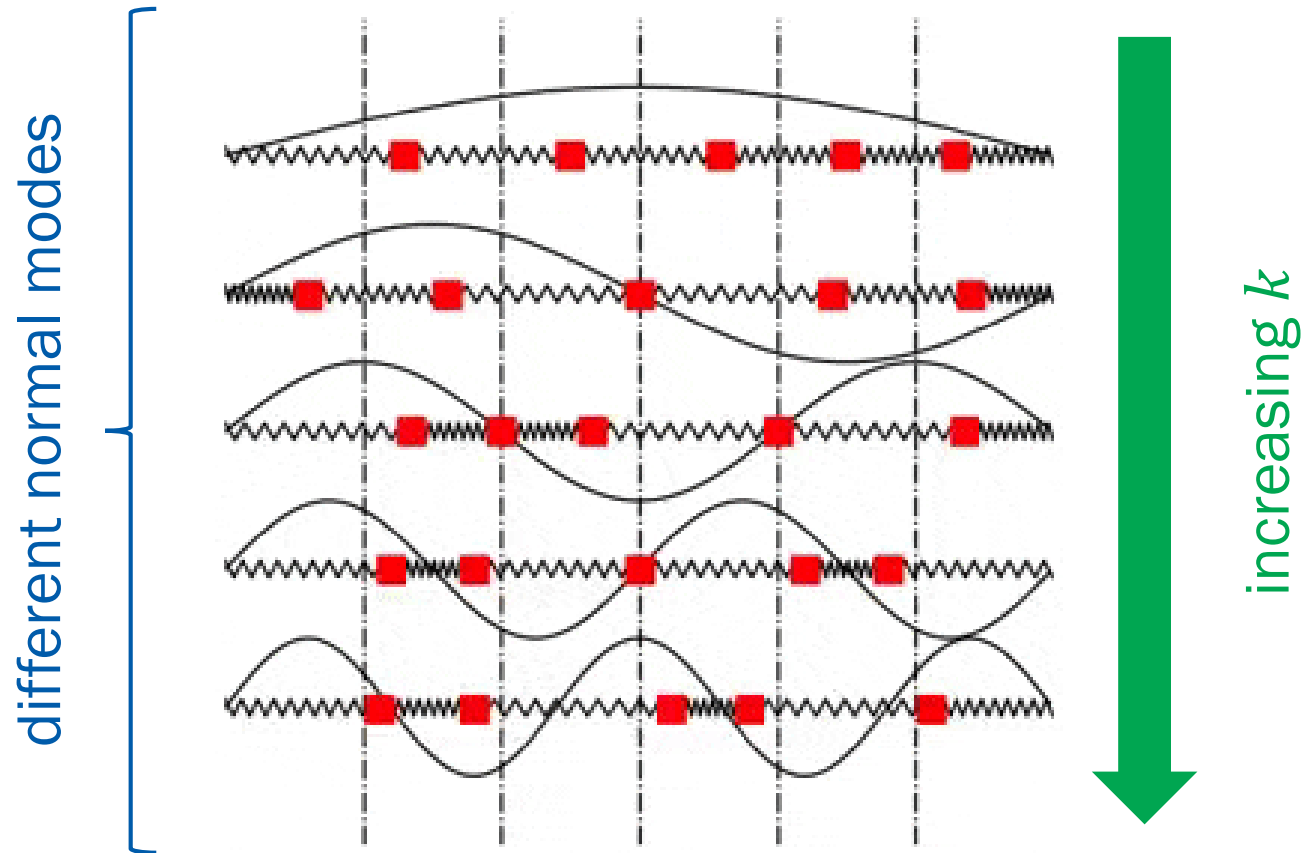
$$m\omega^2 = 4\kappa \sin^2(ka/2)$$

Thus,

$$\omega = 2\sqrt{\frac{\kappa}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|$$

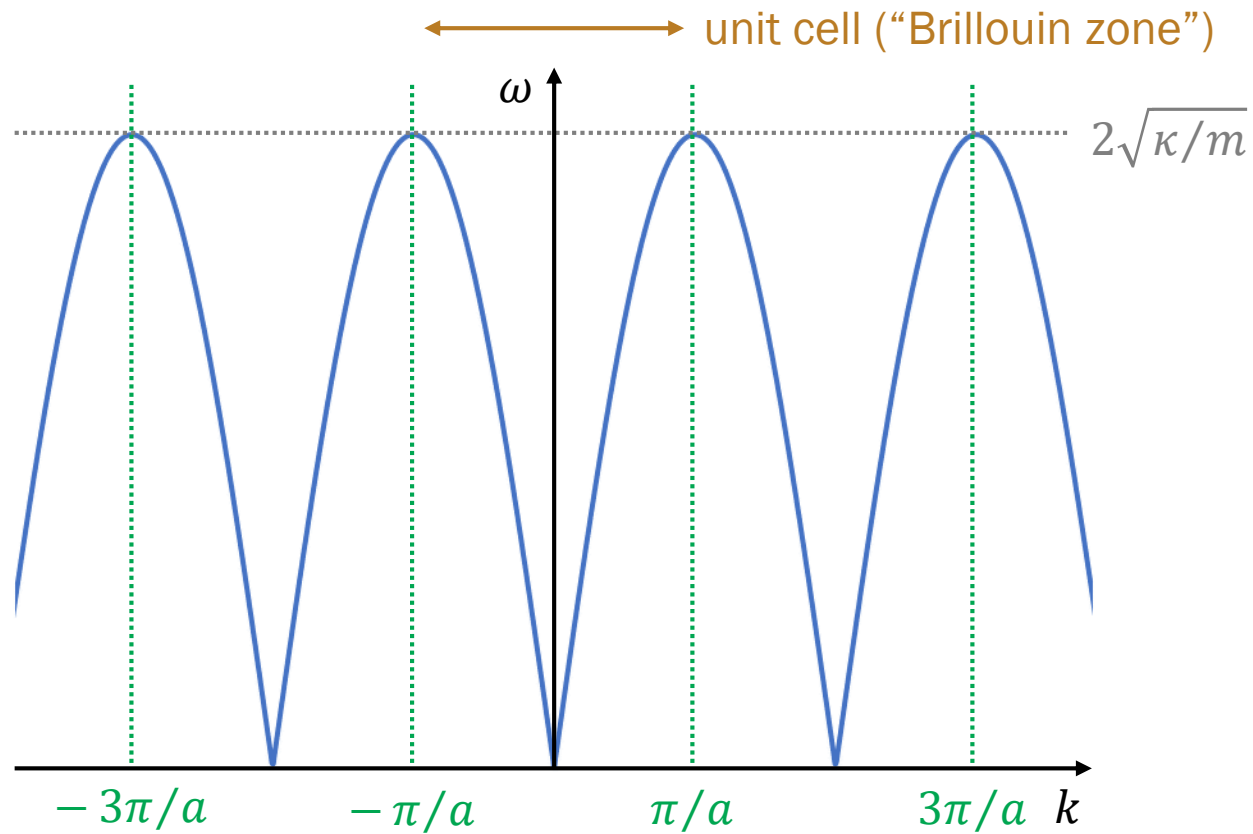
(dispersion relation)

Lattice Vibrations

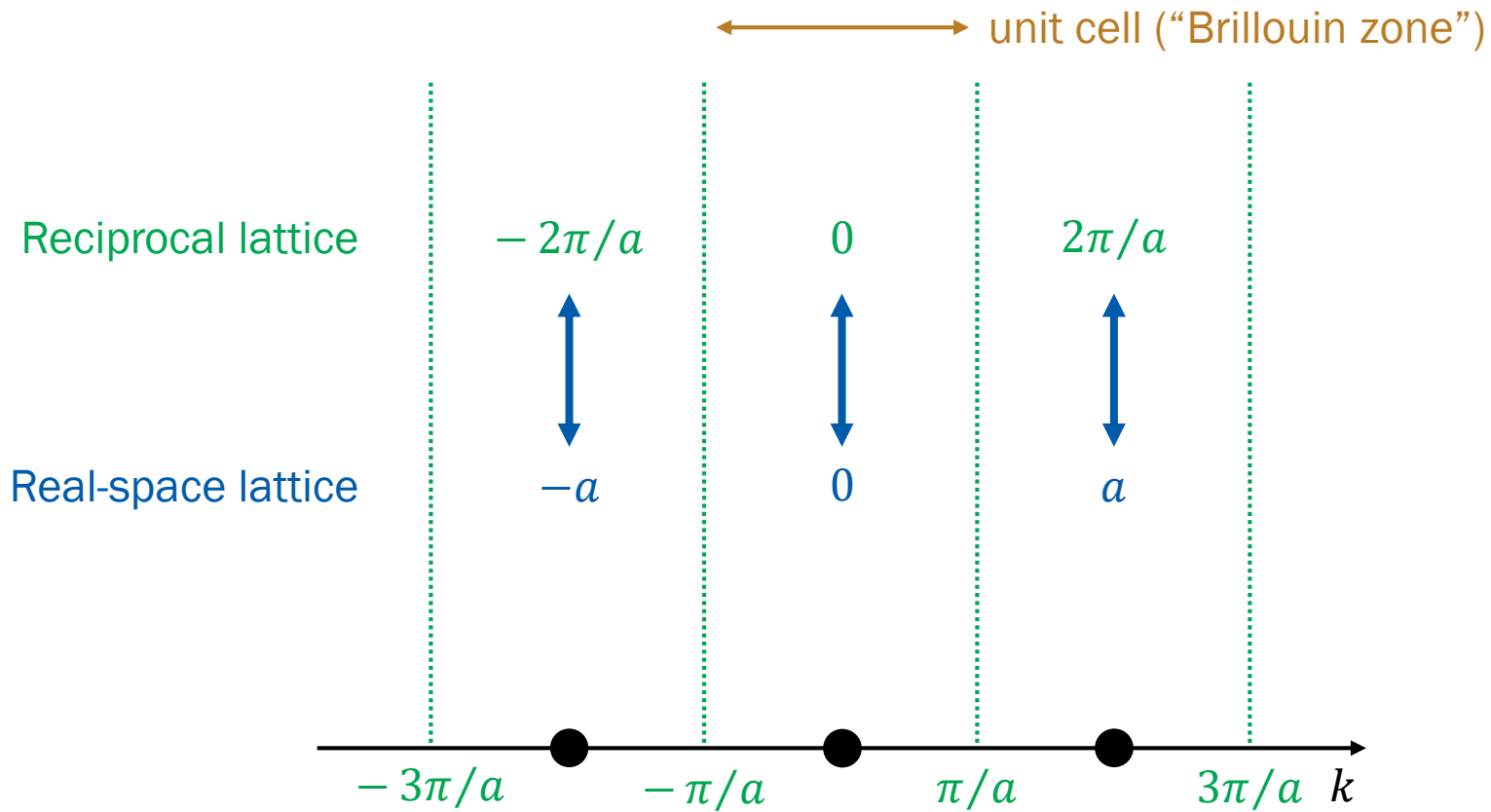


(Image: <https://www.quora.com/What-is-an-intuitive-explanation-of-phonons>)

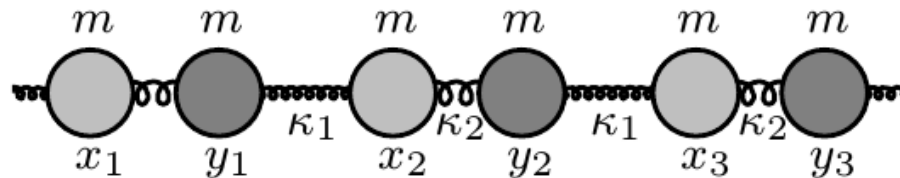
Dispersion Relation



Reciprocal Lattice



Case of Diatomic Solid



Define $q_n = x_n - x_{n0}$ and $q'_n = y_n - y_{n0}$;

$$V_{\text{tot}} = \sum_{i=1}^N \frac{\kappa_1}{2} (x_{i+1} - y_i - a)^2 + \sum_{i=1}^N \frac{\kappa_2}{2} (y_i - x_i - a)^2$$

$$V_{\text{tot}} = \sum_{i=1}^N \left[\frac{\kappa_1}{2} (q_{i+1} - q'_i)^2 + \frac{\kappa_2}{2} (q'_i - q_i)^2 \right]$$

(Image: Simon, *The Oxford Solid State Basics* (2013), Fig. 10.5)

Forces on Two Types of Particles

$$V_{\text{tot}} = \sum_{i=1}^N \left[\frac{\kappa_1}{2} (q_{i+1} - q'_i)^2 + \frac{\kappa_2}{2} (q'_i - q_i)^2 \right]$$

$$F_n = -\frac{\partial V_{\text{tot}}}{\partial x_n} = -\frac{\partial V_{\text{tot}}}{\partial q_n} = -\kappa_1 (q_n - q'_{n-1}) + \kappa_2 (q'_n - q_n)$$

$$F'_n = -\frac{\partial V_{\text{tot}}}{\partial y_n} = -\frac{\partial V_{\text{tot}}}{\partial q'_n} = \kappa_1 (q_{n+1} - q'_n) - \kappa_2 (q'_n - q_n)$$

Newton's Law

$$F_n = -\kappa_1(q_n - q'_{n-1}) + \kappa_2(q'_n - q_n)$$

$$F'_n = \kappa_1(q_{n+1} - q'_n) - \kappa_2(q'_n - q_n)$$

As $F = ma = m(\partial^2 x / \partial t^2)$,

$$m \frac{\partial^2 q_n}{\partial t^2} = -\kappa_1(q_n - q'_{n-1}) + \kappa_2(q'_n - q_n)$$

$$m \frac{\partial^2 q'_n}{\partial t^2} = \kappa_1(q_{n+1} - q'_n) - \kappa_2(q'_n - q_n)$$

Solving the Differential Equations

$$m \frac{\partial^2 q_n}{\partial t^2} = -\kappa_1(q_n - q'_{n-1}) + \kappa_2(q'_n - q_n)$$

$$m \frac{\partial^2 q'_n}{\partial t^2} = \kappa_1(q_{n+1} - q'_n) - \kappa_2(q'_n - q_n)$$

- Ansätze: $q_n = A_x e^{i\omega t - ikna}$ and $q'_n = A_y e^{i\omega t - ikna}$

$$-m\omega^2 A_x = \kappa_2 A_y + \kappa_1 A_y e^{ika} - (\kappa_1 + \kappa_2) A_x$$

$$-m\omega^2 A_y = \kappa_1 A_x e^{-ika} + \kappa_2 A_x - (\kappa_1 + \kappa_2) A_y$$

Eigenvalue Problem

$$-m\omega^2 A_x = \kappa_2 A_y + \kappa_1 A_y e^{ika} - (\kappa_1 + \kappa_2) A_x$$

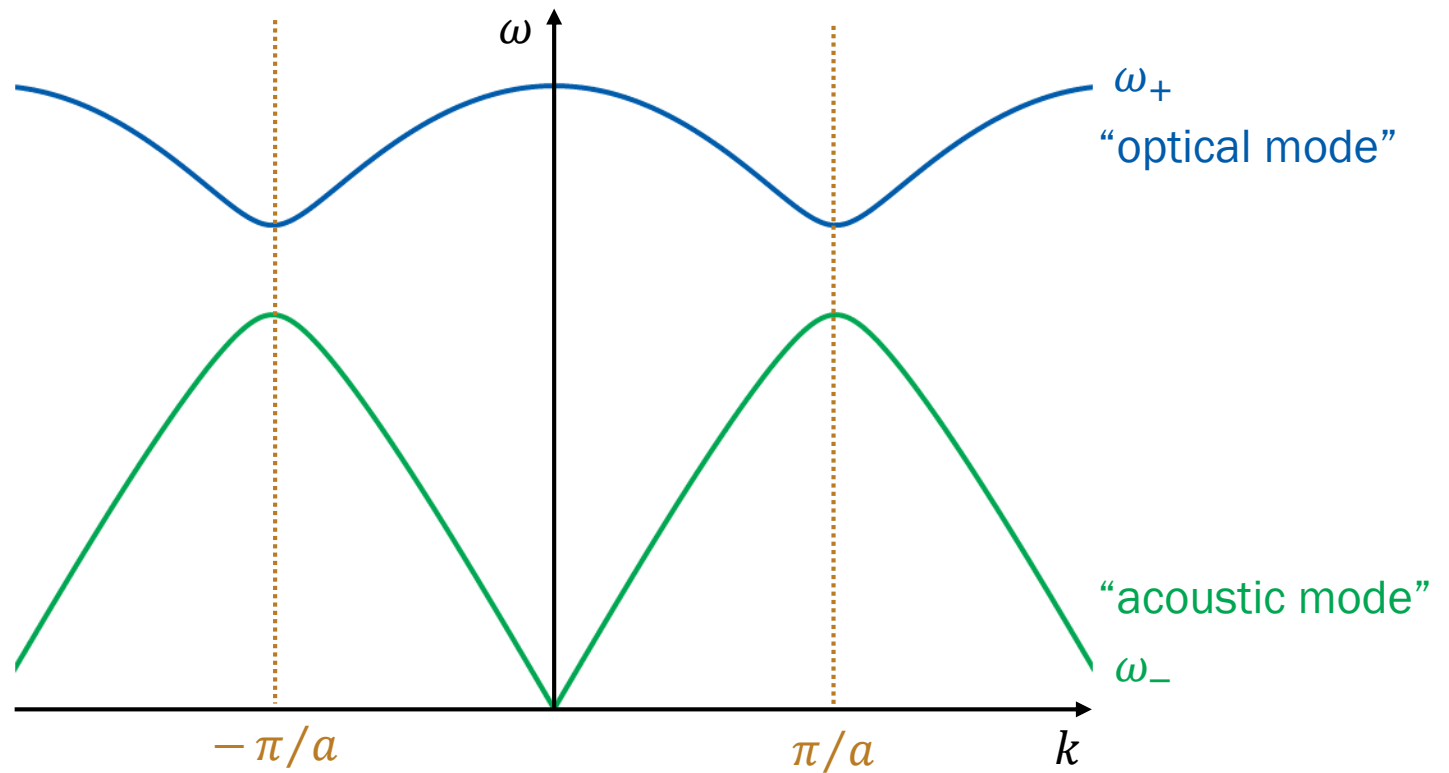
$$-m\omega^2 A_y = \kappa_1 A_x e^{-ika} + \kappa_2 A_x - (\kappa_1 + \kappa_2) A_y$$

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 - \kappa_1 e^{ika} \\ -\kappa_2 - \kappa_1 e^{-ika} & \kappa_1 + \kappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Solving the eigenvalue problem, we obtain

$$\omega_{\pm} = \sqrt{\frac{\kappa_1 + \kappa_2}{m} \pm \frac{1}{m} \sqrt{\kappa_1^2 + \kappa_2^2 + 2\kappa_1\kappa_2 \cos(ka)}}$$

Dispersion Relation



Nature of the Two Modes

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 - \kappa_1 e^{ika} \\ -\kappa_2 - \kappa_1 e^{-ika} & \kappa_1 + \kappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Take $k \rightarrow 0$;

$$m\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} \kappa_1 + \kappa_2 & -\kappa_2 - \kappa_1 \\ -\kappa_2 - \kappa_1 & \kappa_1 + \kappa_2 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

$$\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \frac{\kappa_1 + \kappa_2}{m} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Nature of the Two Modes

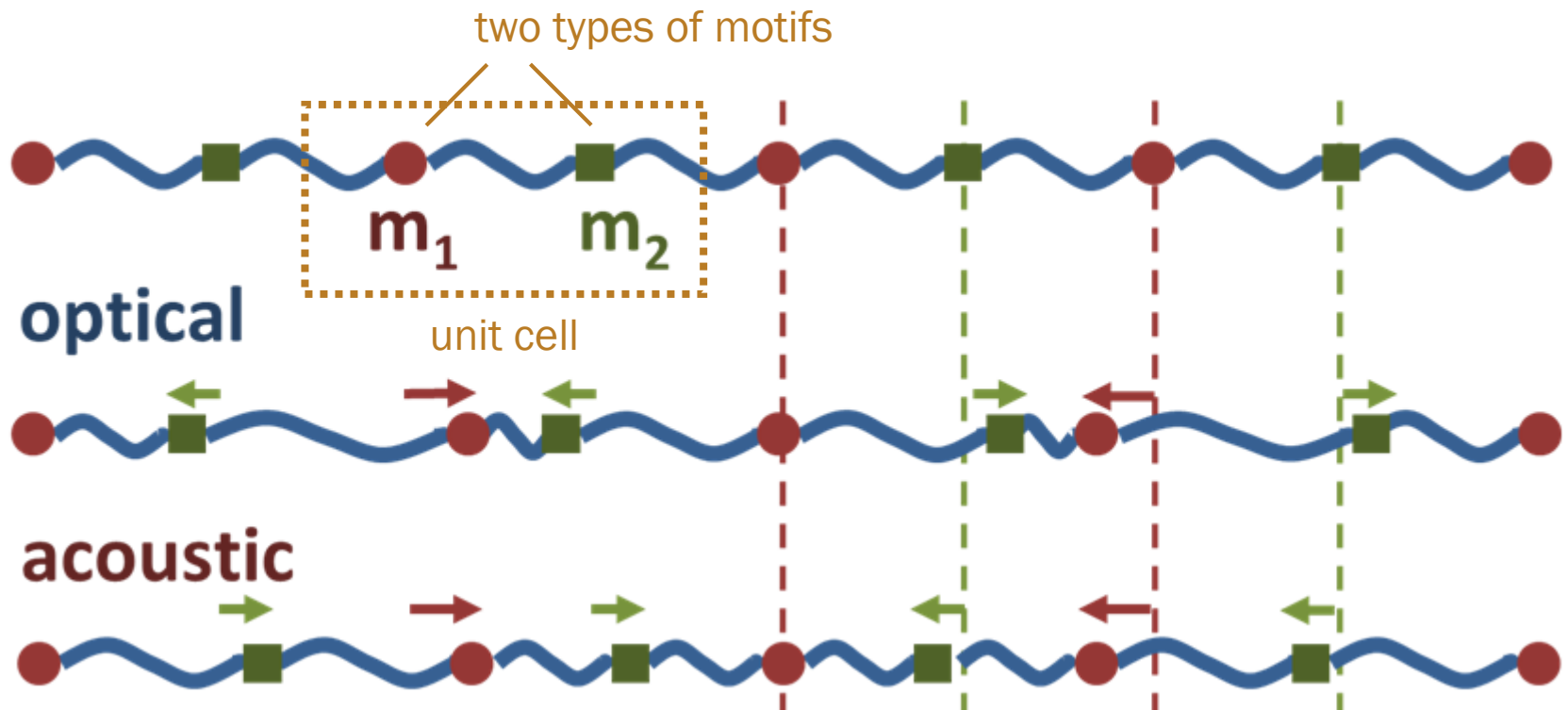
$$\omega^2 \begin{pmatrix} A_x \\ A_y \end{pmatrix} = \frac{\kappa_1 + \kappa_2}{m} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} A_x \\ A_y \end{pmatrix}$$

Solving the eigenvalue problem, the eigenvectors corresponding to ω_+ and ω_- are

$$\begin{pmatrix} A_x \\ A_y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

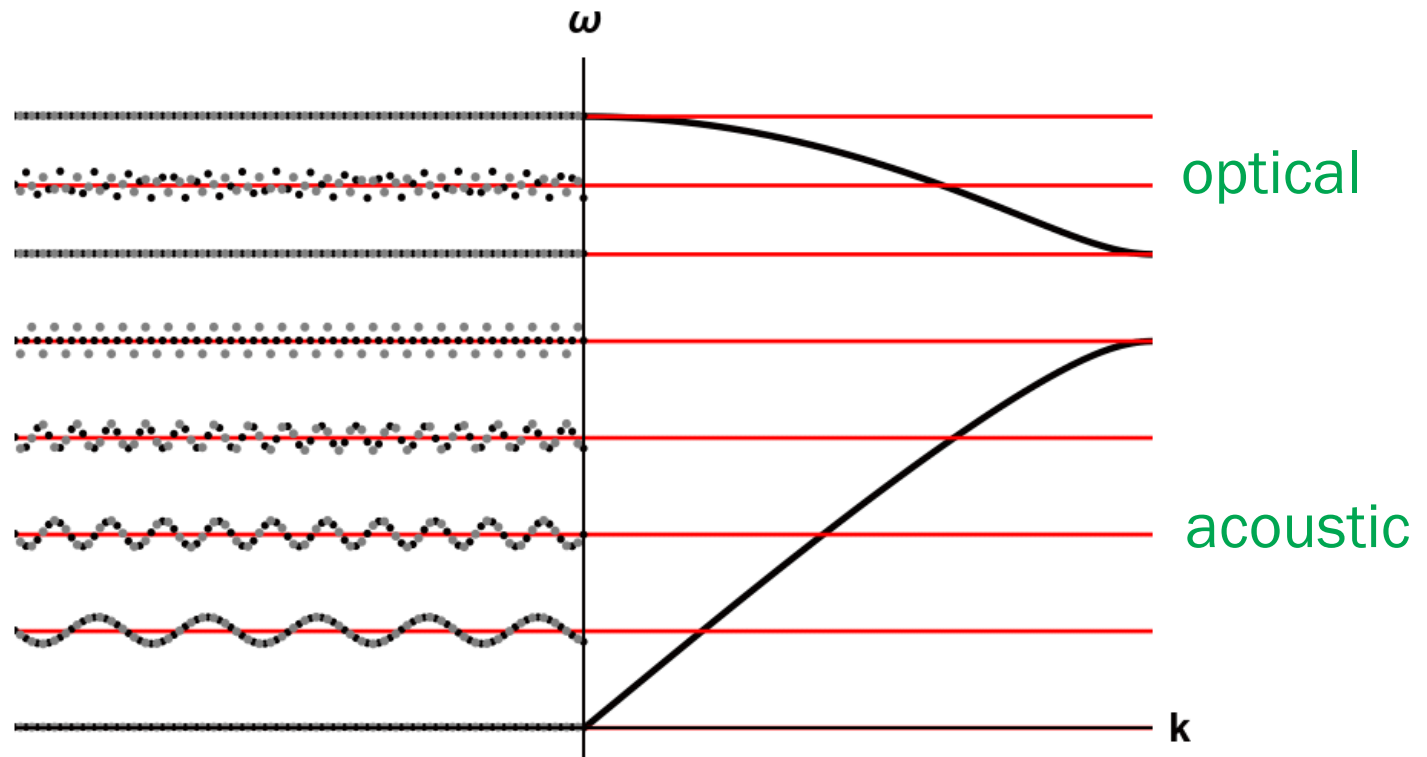
(Recall that $q_n = A_x e^{i\omega t - ikna}$ and $q'_n = A_y e^{i\omega t - ikna}$)

Acoustic and Optical Phonons



(Image: https://commons.wikimedia.org/wiki/File:Optical_%26_acoustic_vibrations.png)

Dispersion Relation

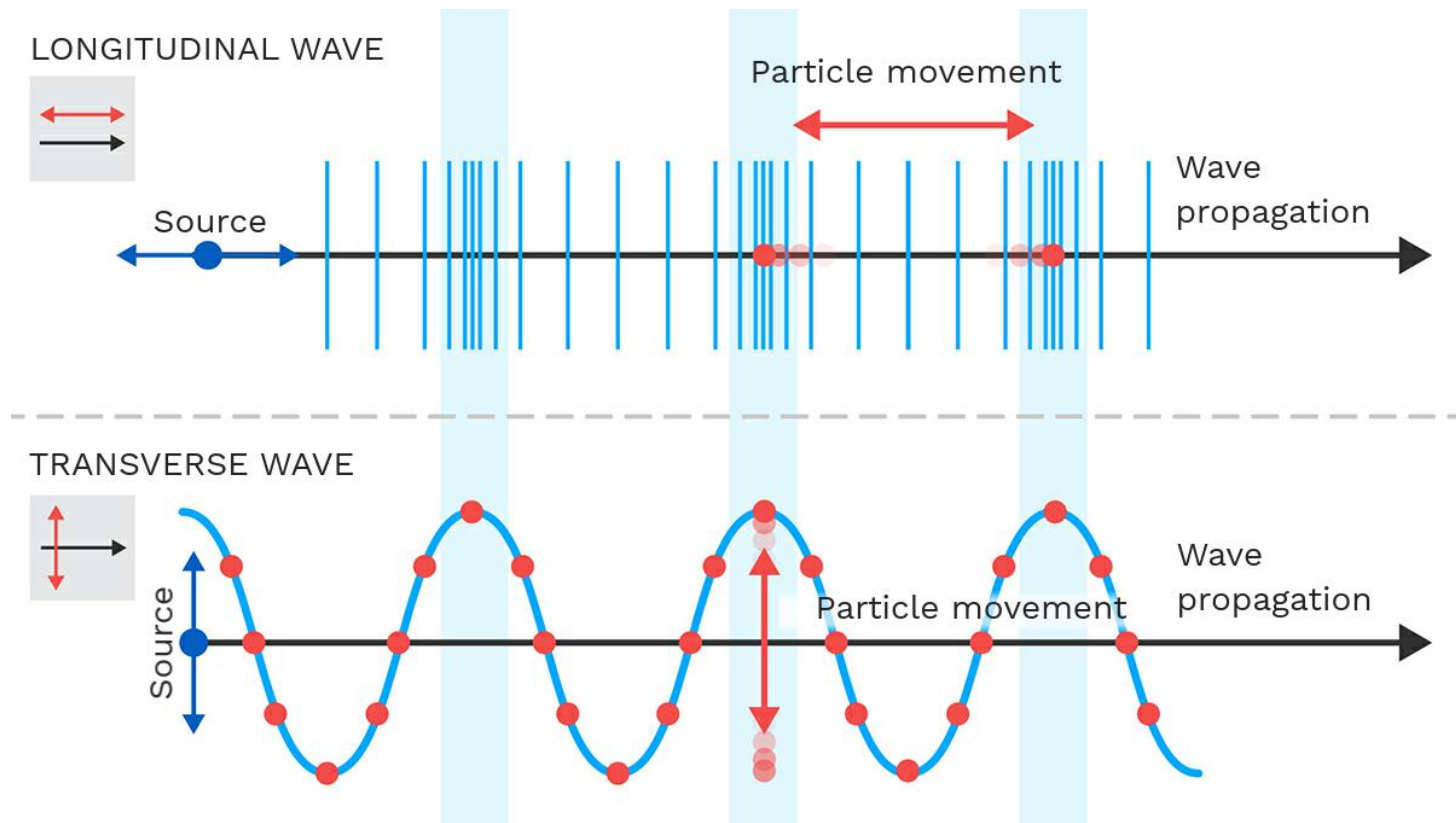


(Image: https://upload.wikimedia.org/wikipedia/commons/d/d7/Diatomic_chain.gif)

Now, Expand to 3D Systems!



Longitudinal vs. Transverse Waves

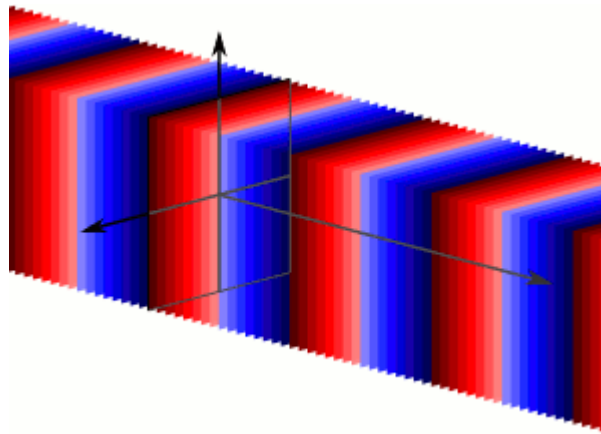


(Image: <https://theory.labster.com/longitudinal-transverse-wave/>)

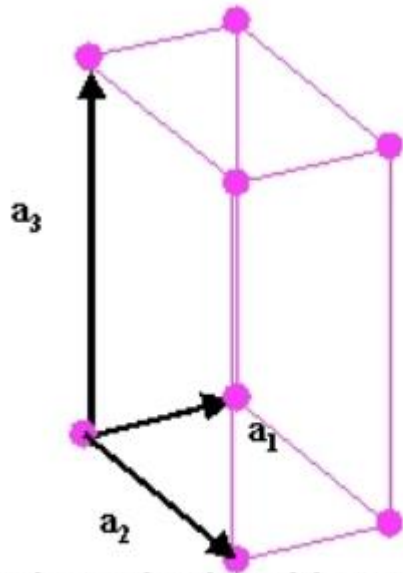
3D Generalization



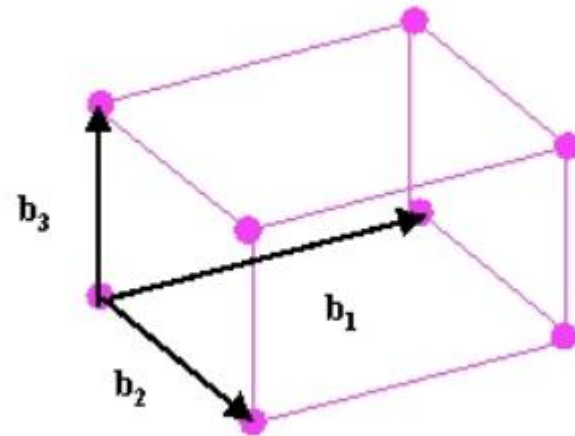
$$A(x, t) = A_0 \exp(ikx - i\omega t)$$



Reciprocal Lattice: Simple Case



Simple Orthorhombic Bravais Lattice
with $a_3 > a_2 > a_1$



Reciprocal Lattice
Note: $b_1 > b_2 > b_3$

Reciprocal Lattice: General Case

For the real-space primitive lattice vectors $\vec{a}_1, \vec{a}_2, \vec{a}_3$, the primitive lattice vectors of the reciprocal lattice must satisfy

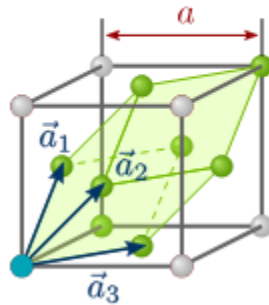
$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$$

which leads to

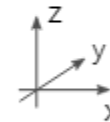
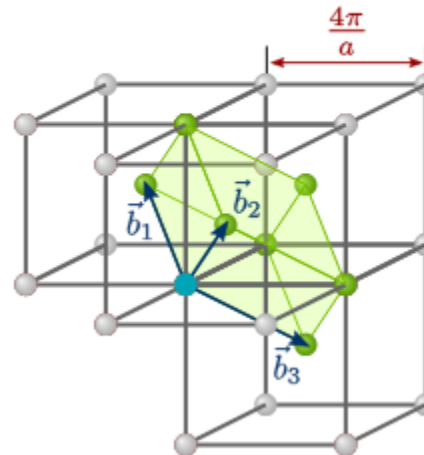
$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}; \quad \vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)};$$
$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

Reciprocal Lattice: Example

direct lattice:
fcc with edge length a

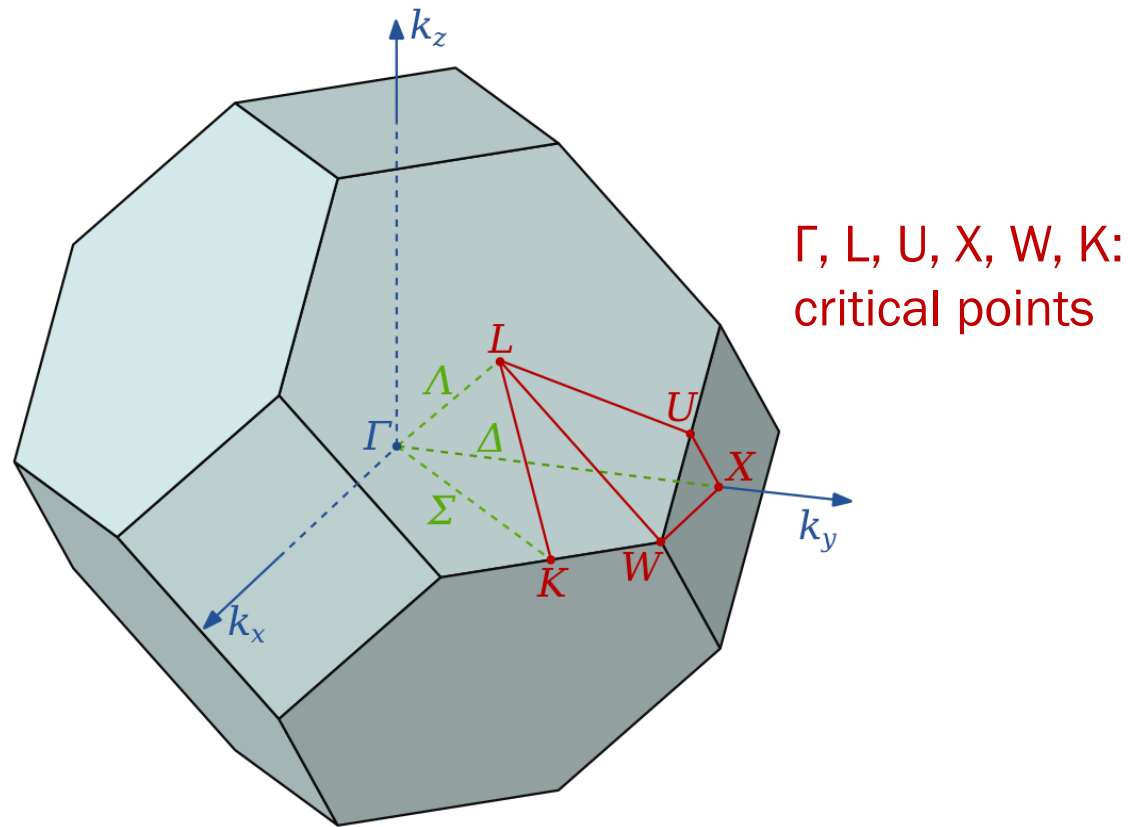


reciprocal lattice:
bcc with edge length $4\pi/a$

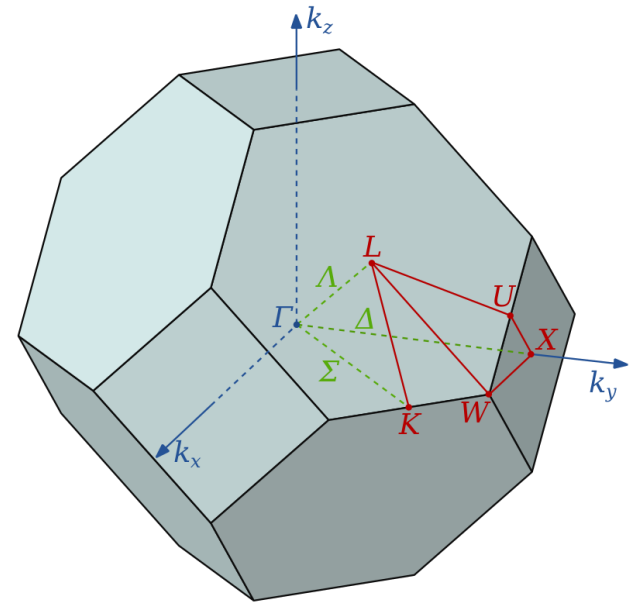
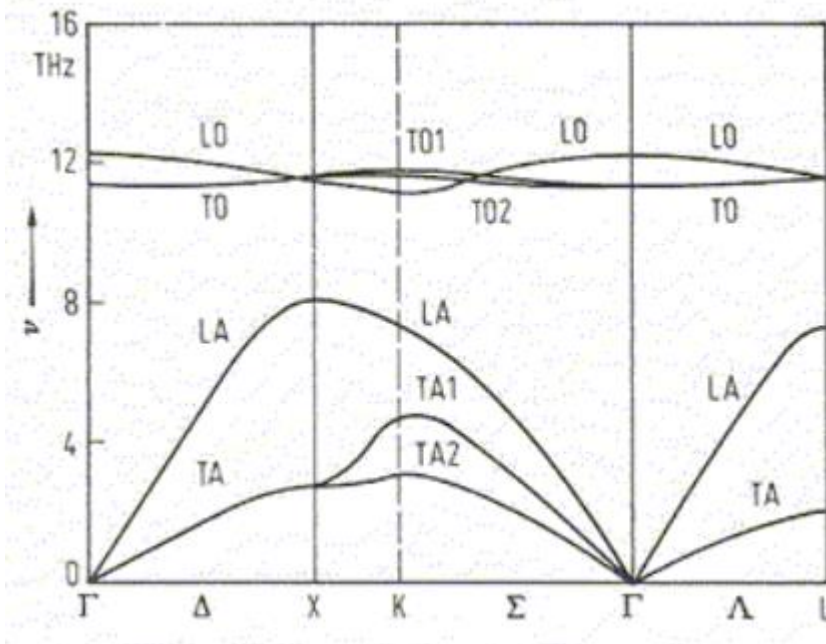


(Image: <https://www.physics-in-a-nutshell.com/article/15/the-reciprocal-lattice>)

Brillouin Zone (FCC)



Dispersion Relation (FCC)



(Images: http://lampz.tugraz.at/~hadley/ss1/problems/problems_phonons.php
[https://commons.wikimedia.org/wiki/File:Brillouin_Zone_\(1st,_FCC\).svg](https://commons.wikimedia.org/wiki/File:Brillouin_Zone_(1st,_FCC).svg))